Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - IX

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Aim

a. **Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.**

Code :

StudentController.cs

```
using Microsoft.AspNetCore.Mvc;
using StudentAPI.Data;
using StudentAPI.Models;

namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;

        public StudentsController(AppDbContext context)
        {
            _context = context;
        }

        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }

        // GET BY ID
        [HttpGet("{id}")]
```

```
public IActionResult GetStudent(int id)
{
    var student = _context.Students.Find(id);

    if (student == null)
        return NotFound();

    return Ok(student);
}

// INSERT
[HttpPost]
public IActionResult AddStudent(Student student)
{
    _context.Students.Add(student);
    _context.SaveChanges();

    return Ok(student);
}

// UPDATE
[HttpPut("{id}")]
public IActionResult UpdateStudent(int id, Student student)
{
    var data = _context.Students.Find(id);

    if (data == null)
        return NotFound();

    data.Name = student.Name;
    data.Age = student.Age;
    data.Course = student.Course;

    _context.SaveChanges();

    return Ok(data);
}

// DELETE
[HttpDelete("{id}")]
public IActionResult DeleteStudent(int id)
{
```

```
            var student = _context.Students.Find(id);

            if (student == null)
                return NotFound();

            _context.Students.Remove(student);
            _context.SaveChanges();

            return Ok();
        }
    }
}
```

Program.cs

```
builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")
    )
);

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI();
```

appSetting.json

```
  "ConnectionStrings": {
    "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"
  }
}
```

Students.cs

```
namespace StudentAPI.Models
{
    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Age { get; set; }
        public string Course { get; set; }
    }
}
```

AppDbContext.cs

```
{
  "name": "sagar",
  "age": 20,
  "course": "IT"
}
```

Curl

```
curl -X 'POST' \
  'https://localhost:44354/api/Students/2' \
  -H 'accept: */*'
```

Request URL

```
https://localhost:44354/api/Students/2
```

Server response

Code	Details
200	Response body

```
{
  "id": 2,
  "name": "sagar",
  "age": 20,
  "course": "IT"
}
```

Download

	Id	Name	Age	Course
▷	2	sagar	20	IT
	NULL	*NULL*	*NULL*	*NULL*

```
using Microsoft.EntityFrameworkCore;
using StudentAPI.Models;

namespace StudentAPI.Data
{
    7 references
    public class AppDbContext : DbContext
    {
        0 references
        public AppDbContext(DbContextOptions<AppDbContext> options)
            : base(options)
        {
        }

        6 references
        public DbSet<Student> Students { get; set; }
    }
}
```

b. Create an API endpoint to calculate Factorial of a Number and test it using Postman.

CODE:-

```
using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {

if (number < 0)
            {
                return BadRequest("Factorial not possible for negative numbers.");
            }
            long factorial = 1;
            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
            return Ok(new
            {
                Number = number,
                Factorial = factorial
            });
        }
    }
}
```

OUTPUT:-